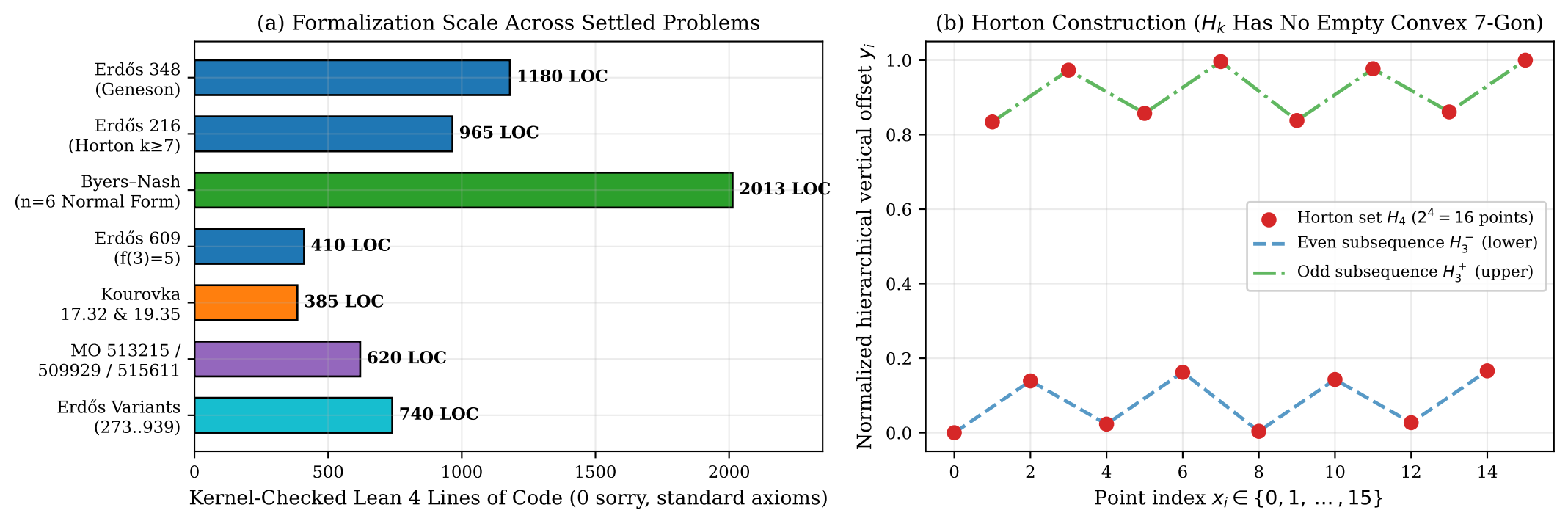
(a) Formalization Scale Across Settled Problems
Erdős 348
(Geneson)
1180 LOC
Erdős 216
(Horton k≥7)
965 LOC
Byers–Nash
(n=6 Normal Form)
2013 LOC
Erdős 609
(f(3)=5)
410 LOC
Kourovka
17.32 & 19.35
385 LOC
MO 513215 /
509929 / 515611
620 LOC
Erdős Variants
(273..939)
740 LOC
0
500
1000
1500
2000
Kernel-Checked Lean 4 Lines of Code (0 sorry, standard axioms)
(b) Horton Construction (H_k Has No Empty Convex 7-Gon)
Normalized hierarchical vertical offset y_i
1.0
0.8
0.6
0.4
0.2
0.0
Horton set H_4 ($2^4 = 16$ points)
Even subsequence H_3^- (lower)
Odd subsequence H_3^+ (upper)
0
2
4
6
8
10
12
14
Point index $x_i \in \{0, 1, \ldots, 15\}$